EXPERIMENT 4: CONVOLUTIONAL NEURAL NETWORK FOR IMAGE CLASSIFICATION

Aim:

To train a CNN on CIFAR-10, evaluate classification accuracy, and visualize learned filters.

Cell 1 – Import libraries and load CIFAR-10

```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt

transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))
])
trainset = torchvision.datasets.CIFAR10(
    root='./data', train=True, download=True, transform=transform
)
testset = torchvision.datasets.CIFAR10(
    root='./data', train=False, download=True, transform=transform
)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=64, shuffle=True)
testloader = torch.utils.data.DataLoader(testset, batch_size=64, shuffle=False)
classes = trainset.classes
```

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Cell 2 – Define CNN architecture

```
class SimpleCNN(nn.Module):
    def __init__(self):
        super(SimpleCNN, self).__init__()
        self.conv1 = nn.Conv2d(3, 16, 3, padding=1)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(16, 32, 3, padding=1)
        self.fc1 = nn.Linear(32 * 8 * 8, 128)
        self.fc2 = nn.Linear(128, 10)
        self.relu = nn.ReLU()

    def forward(self, x):
        x = self.pool(self.relu(self.conv1(x)))
        x = self.pool(self.relu(self.conv2(x)))
        x = x.view(-1, 32 * 8 * 8)
        x = self.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```

Cell 3 – Train the CNN

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = SimpleCNN().to(device)
lossfn = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

for epoch in range(10):
    running_loss = 0.0
    model.train()
    for inputs, labels in trainloader:
        inputs, labels = inputs.to(device), labels.to(device)
```

```
        optimizer.zero_grad()
        outputs = model(inputs)
        loss = lossfn(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item()
    print(f"Epoch {epoch+1}, Loss: {running_loss / len(trainloader):.4f}")
```

```
Epoch 1, Loss: 1.5747
Epoch 2, Loss: 1.2332
Epoch 3, Loss: 1.1037
```

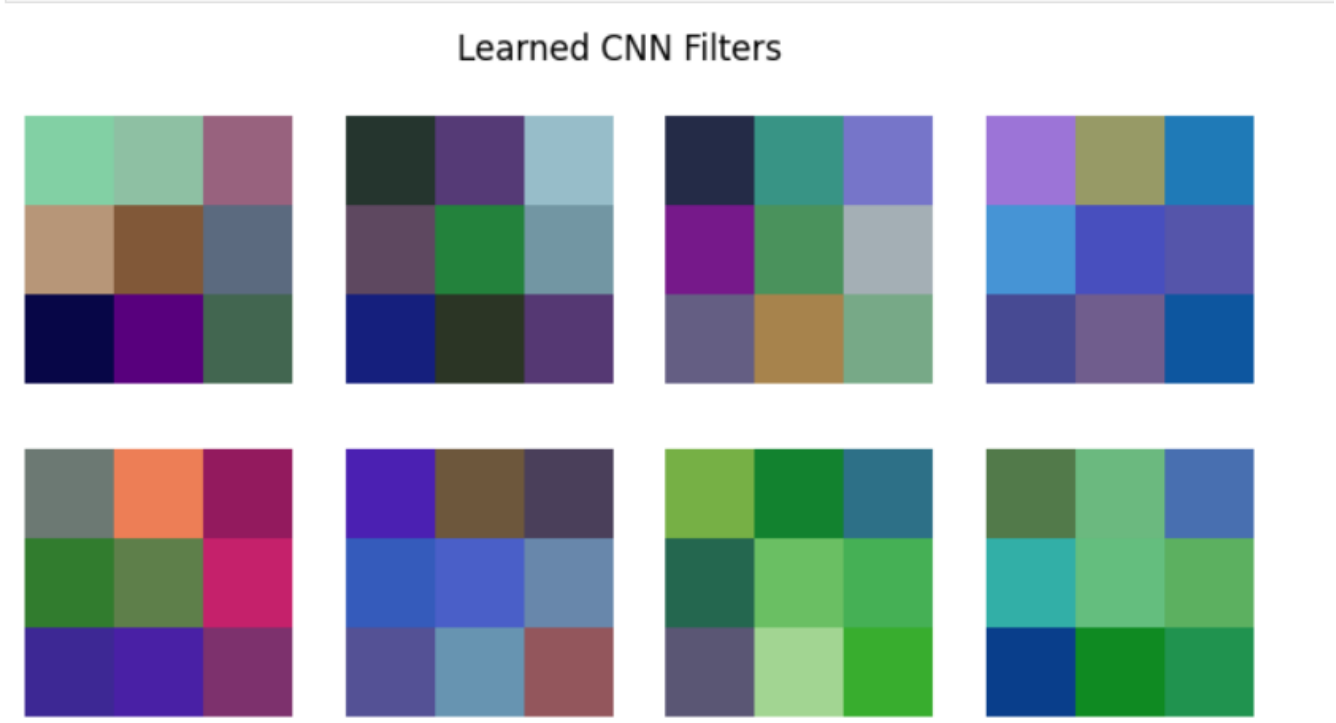
Cell 4 – Evaluate accuracy

```
correct, total = 0, 0
model.eval()
with torch.no_grad():
    for images, labels in testloader:
        images, labels = images.to(device), labels.to(device)
        outputs = model(images)
        _, predicted = outputs.max(1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
print(f"Accuracy on test data: {100 * correct / total:.2f}%")
```

Cell 5 – Visualize learned filters

```
def visualize_filters(layer, n_filters=8):
    filters = layer.weight.data.clone().cpu()
    fig, axs = plt.subplots(1, n_filters, figsize=(15, 4))
    for i in range(n_filters):
        f = filters[i]
        f = (f - f.min()) / (f.max() - f.min() + 1e-8)
        axs[i].imshow(f.permute(1, 2, 0))
        axs[i].axis('off')
        axs[i].set_title(f'Filter {i+1}')
    plt.tight_layout()
    plt.show()

viualize_filters(model.conv1)
```



RESULT

A CNN was trained on CIFAR-10, evaluated on test images, and its learned filters were visualized.